

D589

Chronic and Infectious Diseases

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A norovirus outbreak hits a 120-bed nursing home that recently launched a new dining program. Staff are retrained to wash hands with soap and water between residents. Which link in the chain of infection does this mainly break?

- A. Any method works, since alcohol sanitizer always kills norovirus.
- B. Susceptible host.
- C. Reservoir.
- D. Mode of transmission.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A new respiratory virus, first identified in a busy port city, has a basic reproduction number (R0) of 5. What share of the population must be immune to reach herd immunity?

- A. 5%
- B. 80%
- C. 20%
- D. 50%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

The herd immunity threshold for a disease is 75%. What is its R0?

- A. 0.25
- B. About 1.33
- C. 75
- D. 4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A patient with suspected active pulmonary tuberculosis is admitted to a hospital that was recently re-accredited. Which precautions are standard?

- A. Airborne precautions: a fitted N95 respirator and a negative-pressure room.
- B. Droplet precautions: a surgical mask within close range.
- C. Standard precautions only.
- D. Contact precautions: gown and gloves.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Seasonal influenza spreads mainly through large respiratory droplets. Which precautions fit?

- A. Droplet precautions: a mask for close contact.
- B. Contact precautions only.
- C. Airborne precautions with a negative-pressure room.
- D. No precautions are ever needed for flu.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

After a potluck in a newly renovated church hall, 200 people are interviewed. Of the 120 who ate potato salad, 60 became ill. Of the 80 who did not, 4 became ill. What is the attack rate among those who ate potato salad?

- A. 5%
- B. 32%
- C. 50%
- D. 60

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

In an outbreak, the attack rate was 50% among people who ate potato salad and 5% among those who did not. What is the relative risk?

- A. 10
- B. 32%
- C. 0.1
- D. 45

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A county whose economy depends on three major employers wants to reduce heart disease, stroke, type 2 diabetes, and several cancers at the same time. Which set of shared, modifiable risk factors should it target?

- A. Tobacco use alone.
- B. Workplace policies at the three employers only.
- C. Tobacco use, poor diet, physical inactivity, and excessive alcohol use.
- D. Age, sex, and family history.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A 10-year-old who plays on a travel soccer team has lost weight quickly and is always thirsty. Tests show autoantibodies against insulin-producing cells. Which best describes the underlying process?

- A. Eating too much sugar caused the disease.
- B. Insulin resistance with relative insulin deficiency.
- C. Autoimmune destruction of beta cells, so insulin must be replaced.
- D. It can always be managed with diet alone.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A new rapid test is evaluated at four community clinics. Of 100 people who truly have the disease, 90 test positive. Of 400 people without the disease, 380 test negative. What is the test's sensitivity?

- A. 95%
- B. 10%
- C. About 82%
- D. 90%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A screening test has 95% specificity. Among 1,000 people who do not have the disease, how many will test positive?

- A.** 950
- B.** 50
- C.** 95
- D.** 5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

At an urgent care center that just added weekend hours, an adult with three days of runny nose, sore throat, and cough, and no signs of bacterial infection, asks for antibiotics. What is the best approach?

- A.** Antibiotics kill viruses if the full course is taken.
- B.** Explain that antibiotics do not treat viral colds and can cause harm and resistance; offer symptom care and return precautions.
- C.** Prescribe antibiotics to keep the patient satisfied.
- D.** Prescribe a broad-spectrum antibiotic just in case.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A norovirus outbreak hits a 120-bed nursing home ~~that recently launched a new dining program~~. Staff are retrained to wash hands with soap and water between residents. Which link in the chain of infection does this mainly break?

Target word: link broken **Decides:** Hands carry the virus from person to person.

A	Absolute language	Norovirus resists alcohol sanitizer; soap and water are preferred.
B	Confusion set	Host factors are immunity and health status, not hand hygiene.
C	Adjacent category	The reservoir is where the germ lives, such as infected residents.
D	Correct	Hand hygiene stops the germ from traveling between people.

2. Answer B Concept 2

A new respiratory virus, ~~first identified in a busy port city~~, has a basic reproduction number (R_0) of 5. What share of the population must be immune to reach herd immunity?

Target word: herd immunity threshold **Decides:** $1 - 1/5$.

A	Wrong stem number	Uses R_0 itself as a percent.
B	Correct	$1 - 1/5 = 0.80$.
C	Inverted operation	$1/R_0$ is the part left over, not the threshold.
D	Confusion set	A guess not tied to R_0 .

3. Answer D Concept 2 · transfer

The herd immunity threshold for a disease is 75%. What is its R_0 ?

Target word: solve for R_0 **Decides:** $R_0 = 1 / (1 - 0.75)$.

A	Dropped element	Stops at $1 - 0.75$ without taking the reciprocal.
B	Inverted operation	$1 / 0.75$ uses the threshold instead of $1 - \text{threshold}$.
C	Wrong stem number	The threshold percent.
D	Correct	$1 / 0.25 = 4$.

4. Answer A Concept 3

A patient with suspected active pulmonary tuberculosis is admitted to a hospital ~~that was recently re-accredited~~. Which precautions are standard?

Target word: precaution set **Decides:** TB spreads by airborne particles.

A	Correct	Tiny airborne particles need respirators and air control.
B	Adjacent category	Droplet precautions do not stop airborne particles.
C	Dropped element	Omits the airborne-specific measures.
D	Confusion set	For germs spread by touch, not air.

5. Answer A Concept 3 • transfer

Seasonal influenza spreads mainly through large respiratory droplets. Which precautions fit?

Target word: precaution set **Decides:** Droplet spread.

A	Correct	Droplets travel short distances.
B	Confusion set	Ignores the respiratory route.
C	Inverted operation	Reserved for airborne diseases; routine flu care does not require it.
D	Absolute language	Flu spreads readily in health care settings.

6. Answer C Concept 4

After a potluck in a newly renovated church hall, 200 people are interviewed. Of the 120 who ate potato salad, 60 became ill. Of the 80 who did not, 4 became ill. What is the attack rate among those who ate potato salad?

Target word: attack rate in exposed **Decides:** 60 / 120.

A	Confusion set	4 / 80 is the attack rate among the unexposed.
B	Dropped element	64 / 200 is the overall attack rate, ignoring exposure.
C	Correct	60 ill / 120 exposed = 50%.
D	Wrong stem number	A count, not a rate.

7. Answer A Concept 4 • transfer

In an outbreak, the attack rate was 50% among people who ate potato salad and 5% among those who did not. What is the relative risk?

Target word: relative risk **Decides:** 50% / 5%.

A	Correct	$0.50 / 0.05 = 10$; eaters were ten times as likely to get sick.
B	Confusion set	An overall attack rate, not a comparison.
C	Inverted operation	Divides unexposed by exposed.
D	Adjacent category	$50 - 5$ is the risk difference in percentage points, not relative risk.

8. Answer C Concept 5

A county ~~whose economy depends on three major employers~~ wants to reduce heart disease, stroke, type 2 diabetes, and several cancers at the same time. Which set of shared, modifiable risk factors should it target?

Target word: shared modifiable risks **Decides:** Modifiable and common to all listed diseases.

A	Dropped element	Leaves out diet, activity, and alcohol.
B	Camouflage	A possible setting, not the risk factors themselves.
C	Correct	These four drive much of the listed chronic disease burden.
D	Confusion set	Real risk factors, but not modifiable.

9. Answer C Concept 6

A 10-year-old ~~who plays on a travel soccer team~~ has lost weight quickly and is always thirsty. Tests show autoantibodies against insulin-producing cells. Which best describes the underlying process?

Target word: pathophysiology **Decides:** Autoantibodies point to type 1.

A	Adjacent category	Type 1 is autoimmune, not caused by diet.
B	Confusion set	That describes type 2.
C	Correct	Type 1 causes absolute insulin deficiency.
D	Absolute language	Type 1 requires insulin.

10. Answer D Concept 7

A new rapid test is evaluated ~~at four community clinics~~. Of 100 people who truly have the disease, 90 test positive. Of 400 people without the disease, 380 test negative. What is the test's sensitivity?

Target word: sensitivity **Decides:** True positives / all with disease.

A	Confusion set	380 / 400 is specificity.
B	Inverted operation	The false-negative rate.
C	Adjacent category	90 / (90 + 20) is positive predictive value.
D	Correct	90 / 100 = 90%.

11. Answer B Concept 7 · transfer

A screening test has 95% specificity. Among 1,000 people who do not have the disease, how many will test positive?

Target word: false positives **Decides:** $(1 - 0.95) \times 1,000$.

A	Inverted operation	That is the number of true negatives.
B	Correct	5% of 1,000 are false positives.
C	Wrong stem number	Treats the specificity figure as a count.
D	Dropped element	Uses 5 without applying it to 1,000 people.

12. Answer B Concept 8

At an urgent care center ~~that just added weekend hours~~, an adult with three days of runny nose, sore throat, and cough, and no signs of bacterial infection, asks for antibiotics. What is the best approach?

Target word: stewardship decision **Decides:** Likely viral illness.

A	Confusion set	Antibiotics do not act on viruses.
B	Correct	Avoids unnecessary antibiotics.
C	Adjacent category	Satisfaction does not justify an ineffective drug.
D	Inverted operation	Increases resistance pressure for no benefit.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Chain of infection	Infection needs an agent, reservoir, portal of exit, mode of transmission, portal of entry, and susceptible host; breaking any link stops spread, and hand hygiene breaks the mode of transmission.	Matching an intervention to the wrong link, or assuming alcohol sanitizer works on every germ.	An intervention and "which link does it break?"
2	R0 and herd immunity	Herd immunity threshold = $1 - 1/R0$; so $R0 = 1 / (1 - \text{threshold})$.	Using $1/R0$ as the threshold.	A basic reproduction number or a target immunity level.
3	Transmission-based precautions	Airborne spread (e.g., TB, measles) needs a fitted N95 and negative-pressure room; droplet spread (e.g., influenza) needs a mask for close contact; contact spread needs gown and gloves.	Using a surgical mask for an airborne disease.	The organism's route of spread.
4	Attack rates and relative risk	Attack rate = $\text{ill} \div \text{exposed in a group}$; relative risk = $\text{attack rate in exposed} \div \text{attack rate in unexposed}$.	Using the whole crowd as the denominator, or subtracting when you should divide.	An outbreak table of who ate what and who got sick.
5	Shared modifiable risk factors	Tobacco use, poor diet, physical inactivity, and excessive alcohol use drive much of heart disease, stroke, type 2 diabetes, and several cancers.	Targeting non-modifiable factors or a single factor.	Several chronic diseases and a request for common targets.
6	Type 1 vs. type 2 diabetes	Type 1 is autoimmune destruction of insulin-producing beta cells, so insulin is required; type 2 is insulin resistance with relative deficiency, often managed first with lifestyle and oral drugs.	Calling type 1 insulin resistance or caused by eating sugar.	A child with rapid weight loss and autoantibodies vs. an adult with gradual onset.
7	Sensitivity and specificity	Sensitivity = $\text{true positives} \div \text{all with disease}$; specificity = $\text{true negatives} \div \text{all without disease}$.	Swapping them or confusing sensitivity with positive predictive value.	A 2x2 table of test results against true disease status.
8	Antibiotic stewardship	Antibiotics treat bacterial infections, not viral ones; unnecessary use adds side effects and drives resistance.	Prescribing antibiotics for colds "just in case."	A likely viral illness and a request for antibiotics.

Course scope (public description): Chronic & Infectious Diseases is designed to provide students with a deep understanding of the major diseases that significantly impact public health. Chronic and infectious diseases remain a critical concern globally, and this course aims to empower students with the knowledge and skills needed to address these challenges effectively. Students explore epidemiology, pathophysiology, prevention, and management of various chronic and infectious diseases.